Public Dollar Backing and Voluntary Bank Buffers

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`github.com/carlosgomezp28/ai-project`

Joint IE1 baseline developed with Fabián Méndez Suárez; supervised by Marco Ortiz. Used with Fabián's permission.

Why dollar liquidity is different

Institutional motivation

Dollar reserve requirements are substantially higher than soles requirements because foreign-currency liquidity risk is higher.

This motivates the currency distinction.

It does not prove the buffer comparative static.

BCRP, Inflation Report, March 2024, Box 7.

Research question and scope

How does additional public dollar backing affect a representative bank's optimal voluntary dollar buffer, conditional on aggregate liquidity conditions?

Private decision margin

Voluntary dollar buffer

Aggregate conditions

Fixed across backing scenarios

A conditional private best response.

What is inherited, and what is added?

Joint IE1 baseline

- Two classes; soles and dollars
- Required and voluntary reserves
- Shocks and interbank matching
- U/NU; public dollar coverage
- Endogenous buffers and a future fixed point anticipated

Carlos's individual contribution

- Private cost closure
- Reduced-form cost primitives
- Explicit constrained problem
- Buffer FOC
- Conditional R_D response

Baseline environment: classes, currencies, liquidity

Representative classes and currencies

$$k \in \{L, S\}, \qquad c \in \{s, u\}$$

s: soles; u: dollars

Deposits d_k^c: **amounts**

ρ^c, v_k^c: **ratios to deposits**

Notation mapping: $v_k^c := u_k^c$ in the joint IE1 source.

Signed liquidity shock

$$\omega_k^c = \beta_k A^c + e_k^c$$

Negative flow worsens liquidity; positive flow improves it.

Classes differ in idiosyncratic liquidity-shock dispersion and interbank access.

The buffer is chosen before uncertainty resolves

$$\text{Choose } v_k^c \longrightarrow \text{Shock realizes} \longrightarrow \text{Interbank allocation}$$

$$\longrightarrow \text{Public coverage} \longrightarrow \text{Uncovered deficit}$$

At the time of choice

The bank knows the shock law, reserve regime, private costs, feasible set, and state-contingent coverage rules/schedules.

It does not know realized A^c and e_k^c.

State-contingent coverage schedules are taken as given.

U versus NU: where required reserves can be used

NU: not individually usable

Only the voluntary buffer absorbs the shock.

Required reserves may support collective dollar capacity.

U: usable by the bank

Required and voluntary reserves absorb the shock.

The same principal is not counted again in the fund.

$$F^u = 1_{\{NU\}} \nu_F^u \rho^u D^u + R_D$$

$\nu_F^u \in [0,1]$: mobilizable fraction; D^u: aggregate dollar deposits.

$R_D \geq 0$: additional dollar-backing capacity.

From own liquidity to the uncovered deficit

$$x_k^c = s_k^c / d_k^c = \omega_k^c + v_k^c + \chi_r \rho^c$$

Deficit probability $\pi_k^{c-} \longrightarrow$ Interbank coverage $\Psi_k^{-,c}$

$\longrightarrow$ Public coverage $\phi^c \longrightarrow$ Uncovered deficit

P^c: admitted interbank supply

Q^c: admitted interbank demand

Δ^c: residual deficit amount

$$\Psi_k^{-,c}(A) = \tau_k \min\left\{1, \frac{P^c(A)}{Q^c(A)}\right\}$$

$$\phi^u(A) = \min\left\{1, \frac{F^u}{\Delta^u(A)}\right\}$$

$\chi_{NU} = 0$, $\chi_U = 1$; $\phi^s = 1$ in the elastic soles benchmark. τ_k: contact probability; positive denominators required.

The missing piece is the explicit private closure

Already anticipated in the joint proposal

Endogenous self-insurance and a future aggregate fixed point.

Completed in this individual project

An explicit trade-off between carrying liquidity
and bearing the expected private cost of deficits.

What makes a chosen buffer privately optimal?

The representative bank's conditional private problem

$$\min_{0 \le v_k^c \le \bar{v}_k^c} \underbrace{\kappa_k^c v_k^c}_{\text{buffer carrying cost}} + \underbrace{\mathbb{E}_{A^c}[h_k^c(A^c) L_k^c(v_k^c, A^c)]}_{\text{expected private deficit cost}}$$

V_k^c: cost per unit of deposits; choice: v_k^c.

$$L_k^c(v, A) := \frac{S_k^{c-}(v, A)}{d_k^c}$$

Inherited S_k^{c-} is an amount; new L_k^c is normalized.

New restriction: nonnegative credit on the inherited balance sheet:

$$b_k^c \ge 0 \quad \Longrightarrow \quad \bar{v}_k^c = 1 + \frac{E_k^c}{d_k^c} - \rho^c.$$

Maintain $d_k^c > 0$, $\bar{v}_k^c > 0$, and fixed equity allocation.

What is $h(A)$? Three exclusive deficit-cost components

For this slide only, suppress bank and currency indices; all coverage shares depend on A.

$$h = \Psi^{-} q_I + (1 - \Psi^{-})\left[\phi q_F + (1 - \phi)\ell\right]$$

Interbank-funded

Weight Ψ^{-}

Unit cost q_I

Publicly funded

Weight $(1 - \Psi^{-})\phi$

Unit cost q_F

Uncovered

Weight $(1 - \Psi^{-})(1 - \phi)$

Unit cost ℓ

The mutually exclusive weights sum to one.

Reduced-form costs: $\kappa > 0$; $q_I, q_F, \ell \geq 0$.

Public coverage is not a net-transfer measure.

Why does another buffer unit save π^-?

No liquidity shortfall

An extra unit saves **zero** deficit.

Liquidity shortfall

An extra unit saves **one** unit of deficit, locally.

$$\frac{\partial L_k^c(v, A)}{\partial v} = -\pi_k^{c-}(v, A)$$

Expectation turns the marginal saving into a deficit probability.
For the amount, retain deposit scaling:

$$\frac{\partial S_k^{c-}(v, A)}{\partial v} = -d_k^c \pi_k^{c-}(v, A).$$

The FOC balances marginal cost and avoided deficits

$$\kappa_k^c = \mathbb{E}_{A^c}\left[h_k^c(A^c)\pi_k^{c-}(v_k^{c,*}, A^c)\right]$$

Marginal carrying cost of self-insurance

=

Expected marginal deficit cost avoided

Maintained restriction: $h_k^c(A)$ is independent of this bank's choice.

Domain: an interior optimum.

When does the FOC describe a regular interior optimum?

Conditions

- Feasible and interior: $0 < v^* < \bar{v}$
- Away from support/coverage kinks
- Positive curvature: $\mathcal{H} = V_{vv} > 0$

What supplies curvature?

A positive-probability state with $h > 0$ and a threshold strictly inside the uniform support.

Corners remain possible.

Positive deficit probability alone does not imply positive curvature.

How public dollar backing enters the private incentive

$$R_D \uparrow \implies \phi^u \uparrow \text{ when rationed}$$

$$\implies h \downarrow \text{ if } \ell^u > q_F^u$$

$$\implies \text{marginal benefit of private buffering falls}$$

$$\frac{\partial h_k^u(A)}{\partial R_D} = -\frac{[1 - \Psi_k^{-,u}(A)](\ell^u - q_F^u)}{\Delta^u(A)}$$

Local rationing region: $\Delta^u(A) > 0$, $F^u < \Delta^u(A)$; aggregate schedules fixed.

Candidate result: lower voluntary dollar buffers

$$\frac{dv_k^{u,*}}{dR_D} < 0$$

CONDITIONAL

PRIVATE BEST RESPONSE

NOT A GENERAL-EQUILIBRIUM THEOREM

All strict conditions

- Regular interior optimum; $\ell^u > q_F^u$
- One positive-probability state with *all three*: deficit exposure, $\Psi_k^{-,u} < 1$, and rationed public coverage responsive to R_D
- $P^u(A), Q^u(A), \Delta^u(A)$ fixed across backing scenarios
- Other primitives, shock law, regime, and feasible set also fixed

Why this is not already solved

The general mechanism already exists

Public support affects private liquidity incentives.

Ratnovski (2009); Farhi–Tirole (2012); Freixas–Martin–Skeie (2011); Rochet–Vives (2004).

This project does not claim literature-wide novelty.

The contribution claimed here

Specific private buffer-choice closure and conditional characterization inside the **joint IE1 framework**: two bank classes, soles/dollars, interbank coverage, U/NU, and limited dollar backing.

Detailed proposition/version comparison remains under review.

Plan, evidence, and main risk

Current evidence

28 symbolic checks pass locally. Handwritten derivation completed.

Final-project tasks

- Characterize interior and corner solutions
- Numerical illustrations
- Aggregate buffer fixed point
- Lean formalization of numbered final-paper results

Main risk

Aggregate feedback may overturn the conditional sign, or the optimum may hit a corner or coverage kink.

Report the supported local/conditional result honestly.

Backup

Definitions, derivations, accounting, and literature comparisons.

B1. Complete notation and provenance

Symbol	Meaning	Provenance
$k, \mu_k; c$	Classes L, S, masses; soles/dollars s, u	IE1
d_k, w_k, d_k^c	Deposits and fixed dollar share; $d_k^u = w_k d_k$, $d_k^s = (1 - w_k) d_k$	IE1
E_k^c, b_k^c	Allocated equity and credit amounts; $\sum_c E_k^c = E_k$	IE1
ρ^c, m_k^c, v_k^c	Reserve ratios; $m = \rho + v$, with $v := u$ in source	IE1; rename
ω_k^c, β_k	Signed flow $\omega = \beta A + e$; aggregate-shock loading	IE1
e_k^c, a_k^c	Uniform shock; support half-width $a_k^c > 0$	IE1; condition
A^c, η, p	$A^s = 0$; $A^u = -\eta, 0$ with probabilities $p, 1 - p$	IE1
r, χ_r, x_k^c	U/NU; $\chi_U = 1$, $\chi_{NU} = 0$; $x = s/d$	Shorthand
$s_k^c, S_k^{c-}, S_k^{c+}$	Liquidity position and expected deficit/surplus amounts	IE1
π_k^{c-}, L_k^c	Deficit probability; normalized deficit $L = S^-/d$	IE1; new L
λ_k, τ_k	Access parameter and contact probability $1 - e^{-\lambda_k}$	IE1
$P^c, Q^c, \Psi_k^{-,c}$	Admitted supply/demand; interbank coverage factor	IE1
Δ^c, D^u	Residual aggregate deficit; aggregate dollar deposits	IE1
$F^u, \nu_F^u, R_D, \phi^u$	Dollar capacity, mobilization, additional backing, coverage	IE1
$\phi^s = 1$	Elastic soles benchmark operationalized	Benchmark
$\kappa_k^c, q_l^c, q_F^c, \ell^c, h_k^c$	Private unit-cost primitives and deficit-cost composite	Individual
$V_k^c, \bar{v}_k^c, \mathcal{H}$	Objective; new credit bound; curvature V_{vv}	Individual

Positive denominators and conditional shock law maintained; no-deficit conventions remain open.

B2. Piecewise deficits and probabilities

Fix k, c, A; suppress indices. Let $z = \beta A + v + \chi_r \rho$, $a > 0$.

$$L(v,A) = \begin{cases} -z, & z \le -a, \\ (a-z)^2/(4a), & -a < z < a, \\ 0, & z \ge a, \end{cases} \qquad \pi^-(v,A) = \begin{cases} 1, & z \le -a, \\ (a-z)/(2a), & -a < z < a, \\ 0, & z \ge a. \end{cases}$$

$$L_v = -\pi^-, \qquad L_{vv} = \frac{1}{2a} \quad \text{for } -a < z < a.$$

Curvature is zero outside the support.

Values and first derivatives join; curvature changes at $z = \pm a$.
At a fixed state, $S^- = dL$ and $S^-_v = -d\pi^-$.

B3. Implicit-function derivation

Suppress k; dollars only. All aggregate schedules and other primitives fixed.

$$M(v, R_D) := \mathbb{E}_A[h^u(A; R_D)\pi^{u-}(v, A)], \qquad M(v^*, R_D) = \kappa^u.$$

$$M_v = \mathbb{E}[h\pi_v^-] = -\mathcal{H} < 0, \qquad M_{R_D} = \mathbb{E}[\pi^- h_{R_D}].$$

$$0 = M_v \frac{dv^*}{dR_D} + M_{R_D} \quad \Longrightarrow \quad \boxed{\frac{dv^*}{dR_D} = \frac{\mathbb{E}[\pi^- h_{R_D}]}{\mathcal{H}}}.$$

Strict negativity needs a negative contribution in a positive-probability state.
Regular interior optimum; no support or coverage kink at the point evaluated.

B4. Corners and zero-effect cases

Convex constrained problem

$$V_v = \kappa - \mathbb{E}[h\pi^-]$$

$$v^* = 0: \quad V_v(0) \geq 0$$

$$v^* = \bar{v}: \quad V_v(\bar{v}) \leq 0$$

$$0 < v^* < \bar{v}: \quad V_v(v^*) = 0$$

No local backing contribution if

- $\ell^u = q_F^u$
- $\Psi_k^{-,u} = 1$
- $\pi_k^{u-} = 0$
- Public coverage is saturated

A boundary or kink requires its own argument; the interior derivative cannot be extended through it.

B5. U/NU capacity accounting

NU

$$x_k^c = \omega_k^c + v_k^c$$

$$F^u = \nu_F^u \rho^u D^u + R_D$$

Full mobilization $\nu_F^u = 1$ is a benchmark, not the general case.

U

$$x_k^c = \omega_k^c + v_k^c + \rho^c$$

$$F^u = R_D$$

Required-reserve principal is already usable individually.

$$D^u = \sum_k \mu_k d_k^u, \qquad \Delta^u(A) = \sum_k \mu_k [1 - \Psi_k^{-,u}(A)] S_k^{u-}(A).$$

$$\phi^u(A) = \min\{1, F^u/\Delta^u(A)\}, \qquad \phi^s = 1.$$

Fixed P^u, Q^u, Δ^u is an individual-project restriction; equity allocation and regime also fixed.

B6. Literature: known mechanisms and this closure

Paper	Main mechanism	Already establishes/studies	This project's different scope
Ratnovski (2009)	Precautionary liquidity and expected support	Lender-of-last-resort incentives can distort private liquidity choices	Specific reduced-form IE1 buffer closure
Farhi–Tirole (2012)	Public intervention and collective moral hazard	Expected intervention affects private balance-sheet choices	Fixed aggregates; no systemic or strategic theorem
Freixas–Martin–Skeie (2011)	Interbank allocation and central-bank policy	Liquidity shocks, interbank allocation, and policy response	Private buffer margin under IE1 coverage rules
Rochet–Vives (2004)	Private liquidity failure and lender of last resort	Failures of private provision and public intervention	No coordination-failure theorem

Contribution: closure/application in this two-currency, heterogeneous-bank framework.

Broad comparisons only. Detailed propositions, appendices, versions, and citing papers remain under review; external novelty uncertain.